



Tex C. Lab

Utility for managing encrypted files for Power systems

User Manual

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1 Introduction

Tex C. Lab is a software tool developed to generate encrypted files used by Power systems and their Boot Manager.

The program can generate the files required to:

- Automatically log in to the Boot Manager;
- Change the Administrator user's password;
- Reset the Administrator user's password;
- Update the card parameters;
- Validate new firmware for the card.

The generated files must be transferred to the card using a USB device.

This manual describes the features available in the application and the procedures to follow when using the generated files with the Boot Manager.

2 Requirements and general operation

The Boot Manager manages two access levels:

- **Guest**;
- **Administrator**.

Automatic login and operations related to the Administrator user's password are managed through files generated by **Tex C. Lab**.

To process these files, it is necessary to:

1. Copy the generated file to a USB device;
2. Connect the USB device to the card;
3. Power on or restart the card;
4. Press and hold the **X** button during startup;
5. Enter the **pre-bootloader**;
6. Select the required operation.

2.1 Pre-bootloader

The pre-bootloader is the startup environment accessed by holding the **X** button while powering on the card.

From this environment, it is possible to:

- Access the Boot Manager main menu;
- Load the card parameters;
- Load new firmware.

To process the generated files, it is necessary to go through the pre-bootloader. For login, change, and reset operations, it is sufficient to return to the main menu. For parameters and firmware, the corresponding options must instead be selected.

3 Application interface

The application consists of the following pages:

- **Manual;**
- **About;**
- **Settings;**
- **Generators.**

3.1 Manual page

The **Manual** page redirects to a PDF viewer and automatically opens the user manual.

3.2 About page

The **About** page contains information about the application, including:

- Software name;
- Installed version;
- Information about the software manufacturer.

3.3 Settings page

The **Settings** page allows you to select the destination directory for files generated by the application.

Selecting the output directory

To set the destination directory:

1. Open the **Settings** page;
2. Select the desired directory;
3. Confirm the selection.

All subsequently generated files will be saved in the configured directory.

3.4 Generators page

The **Generators** page contains the generators used to prepare files for use with the Boot Manager.

The available generators are:

- **Generate Login;**
- **Change Password;**
- **Reset Password;**
- **Card Parameters;**
- **Firmware Checksum.**

4 Managing generated files

The files generated by **Tex C. Lab** are divided into two categories.

4.1 Automatically processed files

The following files are automatically processed by the pre-bootloader when they are present on the USB device:

- Login file;
- Password change file;
- Password reset file.

For this reason, these files must be removed from the USB device after use, when they are no longer needed.

Warning: keeping these files on a USB device connected to the card could cause the associated operation to be executed automatically during a subsequent pre-bootloader startup.

4.2 Manually loaded files

The following files are manually selected by the user directly from the pre-bootloader:

- Card parameter file;
- Firmware file.

These files are not processed automatically and can therefore remain on the USB device without causing the operation to be executed unintentionally.

5 Generators

The generators are designed to produce files that are automatically processed by the bootloader, **which is accessed by pressing and holding the controller button.**

5.1 Generate Login

The **Generate Login** generator creates a file that allows automatic login to the Boot Manager using the Administrator credentials.

Required data

- Current Administrator password.

Procedure

1. Open the **Generators** page.
2. Select **Generate Login**.
3. Enter the Administrator password.
4. Start generating the file.
5. Copy the generated file to a USB device.
6. Connect the USB device to the card.
7. Power on or restart the card while holding down the **X** button.
8. Enter the pre-bootloader.
9. Select access to the **Main Menu**.
10. Allow the Boot Manager to process the login file.

5.2 Change Password

The **Change Password** generator creates a file that allows the Administrator password to be changed.

To perform this operation, both the current login file and the new password are required.

Required data

- New Administrator password.

Procedure

1. Open the **Generators** page.
2. Select **Change Password**.
3. Select the current login file.
4. Enter the new password.
5. Start generating the file.
6. Copy the generated file to a USB device.
7. ALSO copy the current login file.
8. Connect the USB device to the card.
9. Power on or restart the card while holding down the **X** button.
10. Enter the pre-bootloader.
11. Select the **Main Menu**.
12. Allow the Boot Manager to process the generated file.

After use, remove both the login file and the password change file from the USB device. If you want to continue using a login file, you must regenerate it using the new password that was just set.

5.3 Reset Password

The **Reset Password** generator creates a temporary file for resetting the Administrator password.

The reset file is sufficient on its own: it is not necessary to add a login file or a password change file.

Required data

- MAC address of the card;
- Expiration date and time of the file.

The MAC address must be entered in one of the formats supported by the application, for example:

- 00:11:22:33:44:55;
- 00-11-22-33-44-55;
- 001122334455

The expiration date and time must be set in the future. The file contains a Unix timestamp used by the Boot Manager to verify its validity.

Procedure

1. Open the **Generators** page.
2. Select **Reset Password**.
3. Enter the correct MAC address of the card.
4. Set a future expiration date and time.
5. Start generating the file.
6. Copy the generated file to a USB device.
7. Connect the USB device to the card.
8. Power on or restart the card while holding down the **X** button.
9. Enter the pre-bootloader.
10. Select the **Main Menu**.
11. Allow the Boot Manager to process the reset file.

The file cannot be used after the specified expiration time and must be removed from the USB device after use.

5.4 Card Parameters

The **Card Parameters** generator prepares the card parameter file for loading through the pre-bootloader.

The parameters must first be downloaded from the Boot Manager. The resulting file, named **POWER.CRD**, can be edited as a text file.

After editing, the file must be processed by **Tex C. Lab**. The program generates the encrypted file:

CARDPARS.ENC

Procedure

1. Download the card parameters from the Boot Manager.
2. Open the **POWER.CRD** file with a text editor.
3. Modify the required parameters.
4. Save the modified file.
5. Open the **Generators** page.
6. Select **Card Parameters**.
7. Select the **POWER.CRD** file.
8. Start generating the file.

9. Copy `CARDPARS.ENC` to a USB device.
10. Connect the USB device to the card.
11. Power on or restart the card while holding down the **X** button.
12. From the pre-bootloader, select **Upload Card Parameters**.
13. Select or confirm the `CARDPARS.ENC` file.
14. Wait for the operation to complete.

The `CARDPARS.ENC` file is manually selected by the pre-bootloader and is not processed automatically. Therefore, if desired, it can be left on the USB device.

5.5 Firmware Checksum

The **Firmware Checksum** generator validates new firmware for loading onto the card by generating the checksum required by the card.

The firmware must therefore first be processed by **Tex C. Lab** before it can be loaded by the Boot Manager.

The program generates the encrypted file:

`CHECKSUM.ENC`

Required data

- Firmware binary file.

Procedure

1. Open the **Generators** page.
2. Select **Firmware Checksum**.
3. Select the firmware binary file.
4. Start generating the file.
5. Copy `CHECKSUM.ENC` to a USB device (together with the firmware itself).
6. Connect the USB device to the card.
7. Power on or restart the card while holding down the **X** button.
8. From the pre-bootloader, select **Upload Firmware**.
9. Wait for the update to complete.

The `CHECKSUM.ENC` file is manually selected by the pre-bootloader and is not processed automatically.

6 Generator summary

Generator		Input	Result and usage
Generate Login		Administrator password	Automatic login file to be used from the Main Menu.
Change Password	Pass-	Current login file and new password	File for changing the Administrator password from the Main Menu.
Reset Password		MAC address and expiration	Temporary file for resetting the password from the Main Menu.
Card Parameters	Parame-	Modified POWER.CRD file	Generates CARDPARS.ENC , to be used with Upload Card Parameters .
Firmware Checksum		Firmware binary file	Generates CHECKSUM.ENC , to be used with Upload Firmware .

7 Warnings

- Always verify that the USB device contains the correct file before connecting it to the card.
- Remove the password change and reset files after use, as they may be processed automatically by the pre-bootloader.
- After changing the password, if you want to use automatic login, remember to regenerate the login file using the new password that was just set.
- Carefully verify the MAC address before generating a reset file.
- Make sure that the reset file's expiration date and time are in the future.
- Do not manually modify encrypted files generated by the application.
- Do not interrupt power to the card during parameter or firmware updates.
- Use only firmware compatible with the card model.

8 Troubleshooting

8.1 The file is not processed

Check the following:

- the file is located in the correct directory on the USB device;
- the file name has not been changed;
- the USB device was connected before startup;

- the **X** button was held down during power-on;
- the correct option was selected in the Boot Manager;
- the file has not expired;
- the file was generated for the correct card.

8.2 The reset file is not valid

Check that:

- the entered MAC address corresponds to the card;
- the MAC address format is correct;
- the card's date and time are correct;
- the file has not passed its expiration time.

8.3 Firmware upload fails

Check that:

- the selected firmware is compatible with the card;
- the file was processed using **Firmware Checksum**;
- the `CHECKSUM.ENC` file was used;
- the operation was started from the **Upload Firmware** option.